College Workshop Syllabus

Course: Applied RAG and Local LLMs
Instructor: Dr. Aisha Raman

Overview
This workshop introduces students to local document intelligence using Retrieval-Augmented Generation (RAG)

Learning objectives
1. Understand how document text is extracted and cleaned.
2. Learn why chunking matters for retrieval and model context.
3. Generate local embeddings with a lightweight sentence-transformer model.
4. Use cosine similarity to find the most relevant chunks.
5. Retrieve context and send it to a local Ollama model.
6. Explain why grounded answers are more reliable than unsupported guesses.

Eligibility requirements
Students should have basic Python knowledge and a willingness to experiment with local AI tools. Prior experien

Assessment policy
The workshop includes a practical lab, a short reflection, and a final demonstration. Students are expected to co

Deadlines
- Registration closes on Friday.
- Lab setup must be completed before the first practical session.
- Demo questions should be submitted at least one day before the final showcase.

Important reminder
The model is not allowed to invent facts. Answers must be grounded in the retrieved document context. If the inf